Fairness Principles as Routing Constraints: How Five Rules for Distributing Urban Air Mobility Noise Respond to Fleet Size

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***Abstract*—The routing objective of an urban air mobility (UAM) network decides which communities bear its noise. We encode five principles of distributive justice (Equal, Sufficientarian, Prioritarian, Utilitarian and Proportional) as hard constraints and lexicographic objectives in one optimising SMT routing solver, and measure how each responds to fleet size: where it routes, how much it congests, what it costs in travel time and how much noise it adds. The instance is a 5×5 map of Austin, TX with three vertiports and a 6 dB cap on added noise. The fleet ladder runs from $N = 6$ to $N = 24$ aircraft, and every cell is solved under four aircraft orderings, so each measurement carries the solver's indifference band. From $N = 14$ every principle holds the most-exposed community at exactly 4.77 dB above ambient in every ordering, where the unconstrained baseline spans 4.77 to 10.00 dB. Every principle is free in travel time to $N = 16$ and costs 3 aircraft-timesteps at $N = 22$ and 24; Equal, Sufficientarian and Prioritarian pay mainly in distance, Utilitarian and Proportional in departure delay. Proportional confines the fleet to 6 of 25 tiles and cuts summed added noise from 37.7 to 19.7 dB at $N = 16$. Every cell is proven optimal in all four orderings; the slowest condition, Proportional, takes a mean of 439 s at $N = 24$.**

***Index Terms*—urban air mobility, fairness, distributive justice, satisfiability modulo theories, multi-agent routing, environmental noise**

I. Introduction

Urban air mobility (UAM) would move passengers across a city in small electric aircraft at low altitude [1], and community noise is among the factors most likely to decide whether such a network may operate [2]. Aircraft fly between fixed vertiports, so the communities beneath the corridors absorb the noise, and the routing objective decides which communities those are. The background is already unequal: noise exposure in the United States, civil aircraft noise included, is higher in lower-income and minority neighbourhoods [3], [4].

Noise-aware routing formulations usually settle the distributional question implicitly, by minimising total exposure or by holding every receptor below a threshold. Political philosophy gives several distinct accounts of a fair division of a burden [5]–[8], and each can be written as a constraint or an objective over the same routing variables. Gao et al. [9] trade efficiency against a parameterised equity criterion in UAM traffic management; Hauptvogel et al. [10] review distributive rules for aircraft noise without committing them to a planner. We fix five named principles, encode each in one optimising SMT solver [11], [12] and compare their routings on one instance. The encoding follows compilation-based multi-agent path finding [13], [14], with one change: the aircraft do not conflict with each other but are coupled through a shared, nonlinear noise externality.

A principle's effect depends on traffic. A cap on added noise constrains nothing while the fleet fits under it, and a detour costs nothing while the direct corridors have room; both change as the fleet grows. Whether a principle changes the routing, what it costs and where it puts the noise are therefore functions of the fleet size N, and a comparison at one N reports one point on each. This paper measures those functions on a 5×5 map of Austin, TX, for five principles and a no-fairness baseline, from $N = 6$ to $N = 24$. Its contributions are three measurements:

1) routing and congestion against N (Section IV-B);
2) travel time and added noise against N, each difference between conditions tested against the solver's indifference among equally optimal routings, and explained through the closed-form capacity a decibel cap grants a tile (Sections IV-C and IV-D);
3) the solve time of each principle across the ladder, every cell proven optimal (Section IV-E).

II. Five Principles

The burden is the noise aircraft add to a community's ambient level, and a principle chooses among routings rather than among arbitrary allocations. We state each principle's general claim and its reading for routing; Table I gives what the solver receives, in the notation of Section III.

Equal: everyone receives the same share [15]; added noise should be as equal as possible across communities. *Sufficientarian*: everyone should reach a threshold, above which differences carry no weight [7]; no community's added noise may exceed a limit. *Prioritarian*: a gain counts for more when it goes to the worst-off [5], [6]; the communities already bearing the most burden should receive the least added noise. *Utilitarian*: total harm should be minimal [16]; the fleet's aggregate burden should be as low as possible. *Proportional*: burden should follow benefit [17], [18]; the communities that gain most from the network should carry more of its noise.

Equal and Sufficientarian are encoded as caps and contribute no objective. Prioritarian needs a protected set P and Proportional a benefit price π_m, both inputs from outside the noise model (Section III-C). Equal is comparative and Sufficientarian absolute, yet under one limit ℓ they coincide on this instance.

TABLE I

CONDITIONS AS THE SOLVER RECEIVES THEM: THE CAPS SENT AND THE TIER-1 OBJECTIVE, MINIMISED BEFORE THE ARRIVAL PENALTY (5). UTILITARIAN AND PROPORTIONAL ARE SHOWN IN THE COUNT FORM THE STUDY RUNS; THEIR DECIBEL FORMS REPLACE $c_{m,t}$ BY $\delta_m(c_{m,t})$. $\ell = 6$ dB; A CAP NOT SENT IS 30 dB, WHICH NO TILE ON THIS MAP CAN REACH.

Condition	Caps sent	Tier-1 objective
Equal	$U{=}\ell$, $D{=}\ell$	none
Sufficientarian	$D{=}\ell$	none
Prioritarian	$D{=}\ell$	$\sum_t \sum_{p\in P} \delta_p(c_{p,t})$
Utilitarian	$D{=}\ell$	$\sum_t \sum_m c_{m,t}$
Proportional	$D{=}\ell$	$\sum_t \sum_m \pi_m\, c_{m,t}$
Baseline	none	none

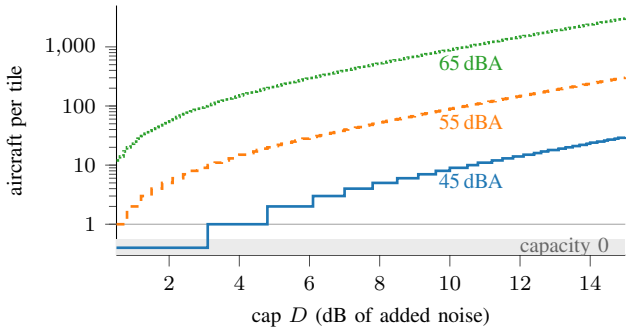


Fig. 1. Aircraft a tile can hold under a cap D on added noise, by ambient level, from (3) with $\alpha = 45$ dBA. At the study's $D = 6$ dB: 2, 29 and 298. Below $D = 3.01$ dB a 45 dBA tile holds none.

Proposition 1. *Let $\delta_{m,t} \geq 0$ be the added noise on tile m at time t, zero on an unoccupied tile. At each t, $\max_m \delta_{m,t} \leq \ell$ implies $\max_m \delta_{m,t} - \min_m \delta_{m,t} \leq \ell$. The converse holds whenever some tile is unoccupied, which is guaranteed when the fleet is smaller than the map, $N < M$.*

Proof. Since $\min_m \delta_{m,t} \geq 0$, the spread is at most the maximum. An unoccupied tile makes the minimum zero and the spread equal to the maximum. □

Every fleet in this study has $N < M = 25$, so the two principles have the same feasible set and the same empty objective.

III. ENCODING AND INSTANCE

A. Routing model

Airspace is $M = 25$ ground tiles $\mathcal{M}$, one per community, tile m carrying ambient level b_m. A fleet of N aircraft is routed over timesteps $t = 0, \ldots, T-1$; aircraft i has origin s_i, destination e_i and position $L_{i,t} \in \mathcal{M}$, with $L_{i,0} = s_i$ and each step a move to one of the eight neighbouring tiles or a hold. An aircraft is audible from its first move through the timestep it lands, marked by the flag $\mathrm{ac}_{i,t}$; an aircraft waiting at its origin or already landed is silent. Tile m then carries

$$c_{m,t} = \sum_i [\, L_{i,t} = m \wedge \mathrm{ac}_{i,t} \,] \tag{1}$$

audible aircraft at time t.

B. Noise and the caps

Aircraft noise adds to ambient noise in energy. With per-aircraft source level α, n simultaneous overflights add

$$\delta_m(n) = 10 \log_{10}\!\left(1 + n\, 10^{(\alpha - b_m)/10}\right) \tag{2}$$

decibels to tile m. δ_m is increasing and concave in n and decreasing in b_m: at $\alpha = 45$ dBA one aircraft adds 3.01 dB to a 45 dBA tile and 0.04 dB to a 65 dBA one. The solver tabulates δ_m at centi-decibel resolution.

The absolute cap D requires $\delta_m(c_{m,t}) \leq D$ for every m and t. Because δ_m is increasing, it is asserted as the count bound $c_{m,t} \leq \kappa_m$ with capacity

$$\kappa_m = \max\{n : \delta_m(n) \leq D\} = \left\lfloor \left(10^{D/10} - 1\right) 10^{(b_m - \alpha)/10} \right\rfloor, \tag{3}$$

up to the rounding of the table. The spread cap U requires $\max_m \delta_m(c_{m,t}) - \min_m \delta_m(c_{m,t}) \leq U$ at every t, encoded with one upper and one lower auxiliary bound per timestep.

Fig. 1 plots (3). At $\alpha = 45$ dBA and $D = 6$ dB a tile holds 2 aircraft at 45 dBA, 29 at 55 dBA and 298 at 65 dBA; at $D = 10$ dB it holds 9, 90 and 900. One cap on added noise is a hundredfold difference in capacity between the quietest and the loudest tier, and it fills the quiet tiles first.

C. Instance and operating point

The instance is a 5×5 grid over Austin, TX, with 8 tiles at 45 dBA, 12 at 55 dBA and 5 at 65 dBA. Vertiports sit at an airport and a downtown core, both 65 dBA, and at an exurban commuter town at 45 dBA. Aircraft fly both directions of the three vertiport pairs: at demand factor $f = 1$, 4 per direction between airport and downtown and 2 on each pair involving the town, $N = 16$. Scaling f from 0.25 to 1.5 gives the ladder $N \in \{6, 8, 14, 16, 22, 24\}$. The source level is $\alpha = 45$ dBA and a timestep is 180 s.

The protected set P is stations 4, 8 and 14, three low-income communities on the vertiport corridors. The Proportional price is

$$\pi_m = \gamma^{\sum_f \rho_f\, \sigma_f(m)}, \tag{4}$$

with base $\gamma = 5$, factor scores $\sigma_f(m) \in [0, 3]$ for income, ambient level, distance to the nearest vertiport and population, and rank weights $\rho = (0.4, 0.3, 0.2, 0.1)$. A low price marks a community that benefits from the network (higher income, louder, nearer a vertiport, less populated), and the objective loads it with traffic.

Every principle faces $D = 6$ dB, so the five share one hard constraint and differ in tier 1. The shipped settings, $D = 10$ dB for Equal and Sufficientarian and a protected set of the three loudest non-vertiport tiles (stations 16, 20 and 25), leave the principles little to act on here: at $N = 16$ no cap fills more than 0.44 of a tile's capacity, and the baseline routing never enters that protected set at any size of the ladder. A screen over the cap and the protected set, four orderings per cell, selected $D = 6$ dB and the corridor set; at $N = 16$ the cap then runs at full capacity under every principle (Section IV-B).

The *baseline* is the same fleet with both caps at 30 dB and no tier-1 objective: the arrival-optimal routing without a principle.

D. Objectives

Each condition minimises two tiers lexicographically. Tier 1 is the principle's objective (Table I). Tier 2, minimised only among tier-1 optima, is the arrival penalty

$$J_{\text{time}} = \sum_i \sum_t [\, \neg \text{ha}_{i,t} \,], \tag{5}$$

the aircraft-timesteps before arrival, where $\text{ha}_{i,t}$ marks aircraft i as landed; time held at the origin counts. Arrival cannot cost a unit of tier 1, and it selects among the routings that tie on it.

The conditions form two lanes. In the *cap lane*, Equal and Sufficientarian have no tier-1 objective, so among cap-feasible routings arrival alone decides. In the *objective lane*, Prioritarian, Utilitarian and Proportional face the same cap and minimise their own objective first. Prioritarian's sums added decibels over P only. Utilitarian's and Proportional's price every audible aircraft-timestep: in count form $\sum_{t,m} c_{m,t}$ is the fleet's airborne time and $\sum_{t,m} \pi_m c_{m,t}$ its price-weighted counterpart, and neither charges an aircraft waiting on the ground. Their decibel forms price $\delta_m(c_{m,t})$ instead. On the shipped settings the decibel forms prove optimality only to $N = 8$, Proportional after 8.2 h; the count forms prove every cell of this study's ladder, and the study runs them.

E. Solving

The horizon starts one timestep above the Chebyshev lower bound on trip length (at the bound for Utilitarian). Phase 1 raises it where the caps need more; phase 2 fixes it and optimises both tiers with Z3's νZ optimiser [11], [12], with arrival made sticky so aircraft can stagger their landings within the horizon.

IV. Results

A. Setup

The six conditions of Table I run at every size of the ladder, from $N = 6$ to 24, on the operating point of Section III-C; the nominal demand, $N = 16$, is marked in every plot. No solver timeout was set, and every cell is proven optimal.

A solve returns one member of a set of equally optimal routings, and which member depends on the order in which the aircraft are listed. Every cell is therefore solved under four orderings of the same fleet. The feasible set and the optimum do not change, so the range over the four is the solver's indifference at that cell, drawn as a band. Ordering k is the same for every condition. A difference between two conditions is *attributable* when their means differ by more than the larger band and their ranges do not overlap; a smaller difference is stated as within the solver's indifference.

B. Experiment 1: routing and congestion

How do each principle's routing and tile occupancy change as the fleet grows? Fig. 2 draws the routings at $N = 8$, 16 and 24, and the left column of Fig. 3 the peak occupancy and the shared tile-timesteps.

Congestion grows with N under every condition: peak aircraft over one tile from 1–3 at $N = 6$ to 11–14 at $N = 24$, and tile-timesteps carrying two or more aircraft from 0–3 to 15–18. On both metrics every principle stays within the solver's indifference of the baseline at every size, with the two exceptions below. The fleet sets how much congestion there is; the principle sets where it sits.

Equal and Sufficientarian pose the same problem at every size of the ladder (Proposition 1), so every difference between them is a difference between members of one optimal set; their arrival penalties agree at every size. At $N = 6$ and 8 the cap is not full, at capacity utilisation 0.50–0.75. From $N = 14$ it binds on the 45 dBA tiles, utilisation 1.00 in every ordering, and from $N = 22$ the corridors widen (Fig. 2, $N = 24$) because a third aircraft may not cross a quiet tile. Sufficientarian carries fewer shared tile-timesteps than the baseline at $N = 16$ (11.75 against 14.25).

Prioritarian holds the protected set at 0.00 dB of added noise at every size in every ordering. Staying off three corridor tiles narrows its footprint to 8–9 tiles against the baseline's 10–12, attributably at $N = 6$, 14 and 16. Its cap binds from $N = 14$ as in the cap lane, and from $N = 22$ it detours around the protected tiles.

Utilitarian's count objective, airborne time, is minimised by flying every aircraft on a shortest path without hovering, which the baseline does at every size. Under the cap Utilitarian attains the same floor at every size, so its tier 1 keeps exactly the cap-feasible routings on shortest paths and arrival chooses among them. Its congestion stays within the baseline's band at every size.

Proportional confines the fleet to the same 6 tiles at every size, the three vertiports and the three 55 dBA tiles between them (stations 9, 12 and 13), against the baseline's 10–12, attributably from $N = 6$ to 22. It loads them: peak occupancy is 3, 5, 8, 10, 12 and 14 aircraft from $N = 6$ to 24, identical in all four orderings, and attributably above the baseline at $N = 8$ (5 against 3.5).

C. Experiment 2: travel time

What does each principle cost against the baseline, in distance and in departure delay? The middle column of Fig. 3 gives both; the arrival penalty (5) counts both.

Through $N = 16$ every principle is free in travel time. At $N = 6$, 8, 14 and 16 each attains the baseline's arrival penalty of 18, 22, 40 and 44 aircraft-timesteps in every ordering, with every aircraft on a shortest path. At $N = 22$ and 24 every principle pays exactly 3 aircraft-timesteps more than the baseline (65 against 62, 69 against 66), with zero band. All five face the same cap and pay the same price for it; they differ in currency.

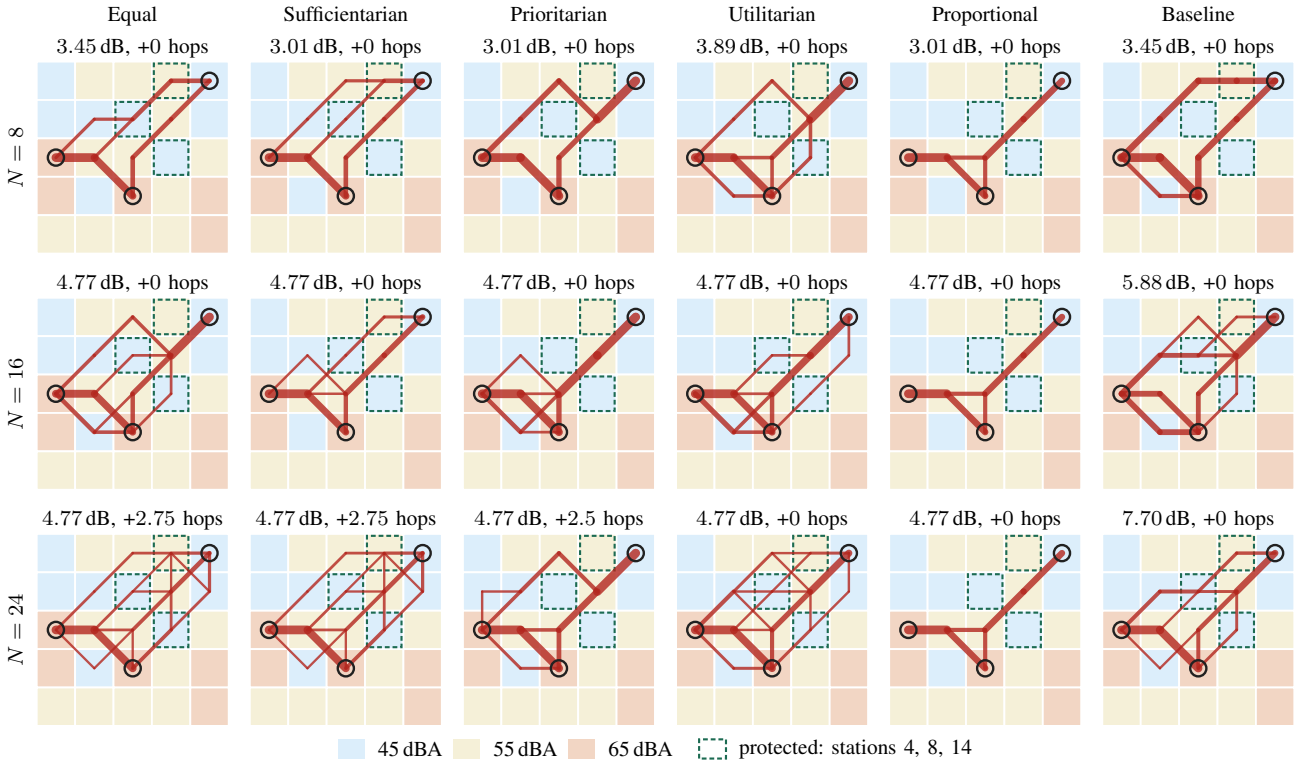


Fig. 2. Routings at $N = 8$, 16 and 24 (rows) under each condition (columns). Tiles are shaded by ambient level, vertiports ringed and Prioritarian's protected set outlined. Each map draws one of the four aircraft orderings. Line width is proportional to the aircraft on each leg summed over the horizon, so two maps can look alike while their routings differ in timing. Titles give peak added noise and extra fleet-hops over the shortest-path floor, averaged over the four orderings.

Equal, Sufficientarian and Prioritarian pay mainly in distance: 2.75, 3.00 and 2.50 extra fleet-hops at $N = 22$ and 2.75, 2.75 and 2.50 at $N = 24$, each attributable. Utilitarian and Proportional fly no extra hop at any size and pay in departure delay: 25 and 27 aircraft-timesteps before first moves at $N = 22$ and 24, against the baseline's 22 and 24, with zero band. The objectives account for the split. Arrival cannot tell a detour from a hold on the ground, and Prioritarian's objective prices only the three protected tiles; Utilitarian's and Proportional's price every airborne aircraft-timestep and none on the ground, so for them a detour costs tier 1 and a hold does not.

D. Experiment 3: added noise and capacity

Where does each principle put the noise as the fleet grows? The right column of Fig. 3 gives the peak and the summed added noise, and Table II the capacity used per ambient tier.

The capacity asymmetry of (3) decides where the cap acts. At $D = 6$ dB a 45 dBA tile holds 2 aircraft and a 65 dBA tile 298, so the cap binds on the quiet tier alone: at $N = 24$ the 45 dBA tiles run at 0.88–1.00 of capacity under every principle and the 65 dBA tiles at 0.02.

Peak. At $N = 6$ and 8 the principles' peaks lie between 3.01 and 3.89 dB, within the baseline's band; one aircraft over a quiet tile adds 3.01 dB. From $N = 14$ every principle returns exactly 4.77 dB in every ordering at every size: two aircraft over a 45 dBA tile, the most (3) admits there. Over $N = 14$, 16, 22 and 24 the baseline's peak averages 5.64, 5.88, 6.70 and 7.70 dB with bands of 2.22, 2.22, 1.76 and 3.98 dB. The binding cap fixes the principles' peak and the aircraft ordering sets the baseline's. The separation clears the band at $N = 22$

TABLE II

CAPACITY UTILISATION BY AMBIENT TIER: PEAK AIRCRAFT OVER A COMMUNITY AS A FRACTION OF THE CAPACITY (3) GRANTS A TILE AT THAT AMBIENT LEVEL, AVERAGED OVER THE TIER'S TILES THAT CARRIED TRAFFIC, FOR THE FIRST AIRCRAFT ORDERING. *max* IS THE LARGEST UTILISATION ON ANY TILE, AVERAGED OVER THE FOUR ORDERINGS. *cap.* IS THE CAPACITY IN AIRCRAFT AT 45/55/65 DBA UNDER THE ROW'S D. EVERY PRINCIPLE SENDS D=6; THE BASELINE SENDS THE WIDE CAP D=30.

N	Condition	D	cap.	45 dBA	55 dBA	65 dBA	max
8	Equal	6	2/29/298	0.50	0.06	0.01	0.62
8	Sufficientarian	6	2/29/298	0.50	0.06	0.01	0.50
8	Prioritarian	6	2/29/298	0.50	0.08	0.01	0.50
8	Utilitarian	6	2/29/298	0.50	0.06	0.01	0.75
8	Proportional	6	2/29/298	0.50	0.11	0.01	0.50
8	Baseline	30	999/9990/99900	0.00	0.00	0.00	0.00
24	Equal	6	2/29/298	0.92	0.19	0.02	1.00
24	Sufficientarian	6	2/29/298	0.92	0.19	0.02	1.00
24	Prioritarian	6	2/29/298	0.88	0.22	0.02	1.00
24	Utilitarian	6	2/29/298	0.92	0.13	0.02	1.00
24	Proportional	6	2/29/298	1.00	0.32	0.02	1.00
24	Baseline	30	999/9990/99900	0.00	0.00	0.00	0.01

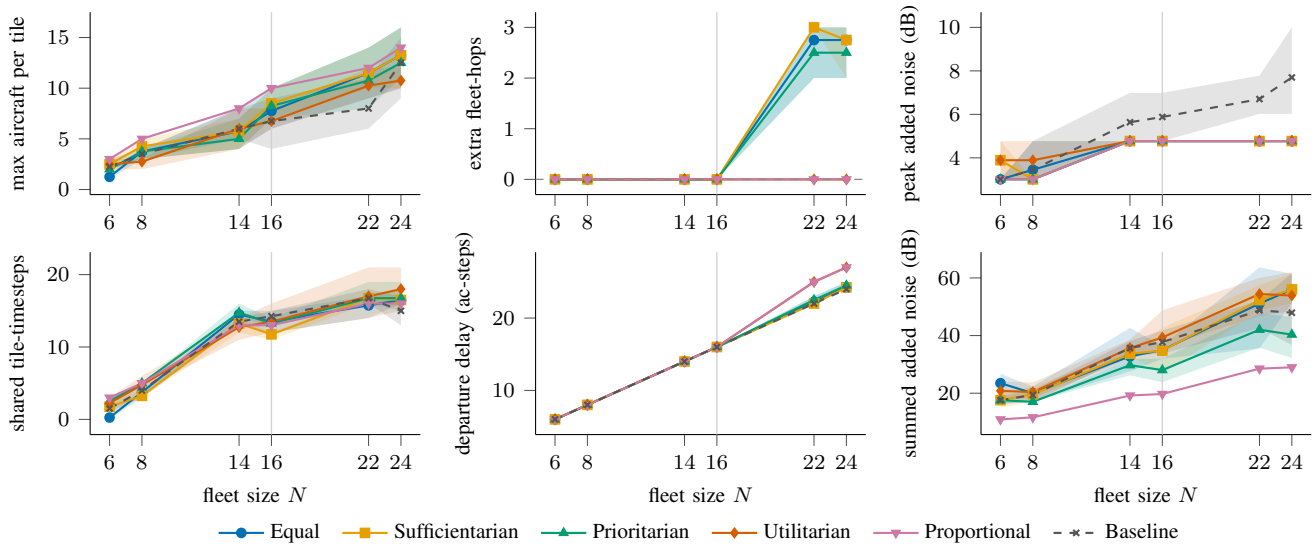


Fig. 3. Congestion (left), travel time (middle) and added noise (right) against fleet size. Lines join the mean over four aircraft orderings; the band is the range over the orderings, the solver's indifference at that cell. Departure delay counts aircraft-timesteps before each aircraft's first move, the origin step included, so a fleet departing at once scores N. Summed added noise adds $\delta_m(c_{m,t})$ over tiles and timesteps. The vertical line marks the nominal demand, $N = 16$.

for all five principles.

Summed noise. Proportional's summed added noise has zero band at every size: 10.9, 11.6, 19.2, 19.7, 28.5 and 29.0 dB from $N = 6$ to 24, against the baseline's 17.6, 19.4, 35.7, 37.7, 48.8 and 47.9 dB. The difference is attributable at every size but $N = 24$, where the baseline's band is 24.9 dB wide. Proportional occupies 13 tile-timesteps at $N \leq 16$ against the baseline's 16–19 with the same airborne time, and since δ_m is concave, fewer occupied tile-timesteps add less in sum. Prioritarian's summed noise is attributably lower at $N = 14$ and 16 (29.7 and 28.0 against 35.7 and 37.7 dB). Equal, Sufficientarian and Utilitarian stay within the baseline's band at every size.

E. Experiment 4: solver performance

How does solve time grow with the fleet? Fig. 4 gives the mean wall time to a proven optimum over the four orderings, with eight solves running in parallel on one 14-core workstation. Every condition proves every cell. Equal, Sufficientarian and Prioritarian solve within a factor of two of the baseline at every size, and the baseline takes 5.9 s at $N = 6$ and 200 s at $N = 24$. Utilitarian is the fastest condition through $N = 16$, at 1.6 s for $N = 6$ and 8.5 s for $N = 16$ against the baseline's 5.9 and 55.3 s. Proportional is the slowest at $N = 22$ and 24, at 227 and 439 s.

F. Summary

Table III collects each principle's response to fleet size.

V. Conclusion

We encoded five distributive principles as caps and lexicographic objectives in one routing solver and measured their response to fleet size on one instance, from $N = 6$ to $N = 24$, every cell proven optimal under four aircraft orderings.

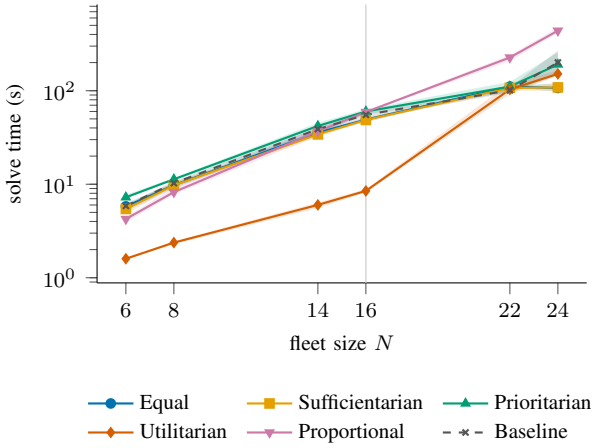


Fig. 4. Mean wall time to a proven optimum over four orderings, eight solves in parallel; the band is the range over the orderings.

Equal and Sufficientarian pose one problem on this ladder: they keep congestion within the solver's indifference of the baseline at all but one cell, fix the peak at 4.77 dB once the cap binds at $N = 14$, and pay 2.75 to 3.00 extra fleet-hops for it from $N = 22$. Prioritarian holds its protected communities at 0 dB at every size, narrows the fleet's footprint, lowers summed noise at $N = 14$ and 16, and pays in distance from $N = 22$. Utilitarian, in count form, keeps every aircraft on a shortest path at every size and pays the same price as departure delay. Proportional confines the fleet to six tiles with zero band at every size, holds summed added noise below the baseline's from $N = 6$ to 22, and also pays on the ground. The baseline's peak varies with the aircraft ordering alone, by up to 3.98 dB.

For a designer, fleet size decides what the choice of principle

TABLE III

RESPONSE OF EACH PRINCIPLE TO FLEET SIZE: THE CHANGE AGAINST THE BASELINE AND THE SIZES N AT WHICH IT HOLDS. "AS BASELINE" IS WITHIN THE SOLVER'S INDIFFERENCE AT EVERY SIZE, EXCEPT SUFFICIENTARIAN'S FEWER SHARED TILE-TIMESTEPS AT $N = 16$. TRAVEL TIME IS FREE TO $N = 16$ FOR ALL FIVE AND COSTS 3 AIRCRAFT-TIMESTEPS AT $N = 22$ AND 24; THE COLUMN GIVES THE CURRENCY. THE PEAK IS FIXED AT 4.77 dB WITH ZERO BAND FROM $N = 14$. PRIORITARIAN'S NARROWER FOOTPRINT IS ATTRIBUTABLE AT $N = 6$, 14 AND 16. *s* IS THE MEAN WALL TIME TO A PROVEN OPTIMUM AT $N = 24$.

	Congestion	Travel time	Added noise	s
Equal	as baseline	distance, 22+	peak fixed, 14+	107
Sufficientarian	as baseline	distance, 22+	peak fixed, 14+	109
Prioritarian	8–9 tiles	distance, 22+	P at 0 dB; peak 14+	191
Utilitarian	as baseline	delay, 22+	peak fixed, 14+	151
Proportional	6 tiles, 6–22	delay, 22+	sum 6–22; peak 14+	439
Baseline	–	–	set by ordering	200

buys. Below the size at which the cap fills, Equal, Sufficientarian and Utilitarian return routings within the solver's indifference of the baseline, and only the objectives of Prioritarian and Proportional move the routing, onto fewer tiles. Above it, every principle fixes the peak and pays the same arrival price, and the choice is which communities carry the remaining noise and whether the delay is taken in the air or on the ground. The cap, shared by all five, sets the peak and the size at which travel starts to cost; the principle sets where the noise goes and the currency of the cost. In count form every principle proves optimal across the ladder, the slowest in a mean of 439 s at $N = 24$.

The acoustic model is coarse, with uniform tiles and point sources without altitude or directivity [2], and the findings rest on one instance and one cap. Four orderings bound the solver's indifference from below.

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